R13

Code No: 118BJ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2019
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) Write major levels of abstraction in the design process of embedded system. [2]
b) List out the features of 8051. [3]
c) Classify the instruction set of the 8051. [2]
d) Discuss the advantages of assembly language. [3]
e) Write about semaphore. [2]
f) Define mailbox and pipe [3]
g) Write the laboratory tools of the embedded system [2]
h) Differentiate the host machine with the target machine. [3]
i) How to organize the memory? [2]
j) Classify the features of SHARC. [3]

PART - B

(50 Marks)

2.a) Explain in detail about the embedded system design process.
b) Explain the memory organization of 8051 in detail. [5+5]

OR

3.a) Draw block diagram of 8051 and explain each block in detail.
b) Discuss about complex systems and microprocessor. [6+4]

4.a) Briefly explain various programming tools and techniques.
b) Explain the Logical Instructions in 8051 with examples. [5+5]

OR

5.a) Explain how subroutines calls are executed in 8051.
b) List and explain the arithmetic group instructions of 8051 microcontroller. [5+5]

6.a) Explain keyboard interface with microcontroller system and keyboard configurations.
b) Explain in detail about multiple interrupts and how to priorities them? [5+5]

OR

7.a) Write short notes on the following:
i) Message queues
ii) Timer Functions
b) Write a detailed note on Task, task states with the help of an example. [5+5]

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8.a) What are the several ways of getting software file into target system? Explain them.
b) Explain about encapsulating semaphores and queues in detail. [5+5]

OR

9.a) Explain the hard real-time scheduling considerations.
b) Explain the linker and locators for embedded software? [5+5]

10.a) Explain the protocols of the embedded network system.
b) Explain theory of operation and requirements in designing an Elevator controller. [5+5]

OR

11.a) Discuss about ARM with block diagram.
b) Explain I2C bus structure with the help of bus architecture, state transmission graph, byte format. [5+5]

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